[AGRI] AI GOVERNANCE RISK INTELLIGENCE PLATFORM

Customer Churn Prediction Model v3.0

Retail Banking — Customer Analytics · 2LoD Assessment · May 10, 2026

ELEVATED

&' ESCALATION REQUIRED

EXECUTIVE SUMMARY

The system demonstrates strong foundational controls with a clean validation opinion and transparent methodology; however, the reliance on logistic regression for churn prediction in a dynamic retail banking environment presents inherent model risk due to potential non-linear pattern capture failures. While the human-in-the-loop design mitigates direct automated harm, the absence of documented bias testing on the new engagement features and lack of explicit monitoring for indirect discrimination elevate the residual risk profile.

Escalation Rationale

Escalation is required due to the ELEVATED risk in Bias and Fairness and Incident/Drift dimensions. The introduction of new features without documented bias testing and the lack of recent drift analysis post-retraining represent control gaps that could lead to undetected model degradation or discriminatory outcomes, requiring 1LoD remediation before the next scheduled review.

RISK DIMENSION RATINGS

Dimension	Rating	Findings
DATA GOVERNANCE	LOW	3 finding(s) · 2 gap(s)
MODEL EXPLAINABILITY	LOW	3 finding(s) · 1 gap(s)
BIAS & FAIRNESS	ELEVATED	2 finding(s) · 2 gap(s)
HUMAN OVERSIGHT	LOW	3 finding(s) · 1 gap(s)
REGULATORY ALIGNMENT	LOW	3 finding(s) · 1 gap(s)
INCIDENT & DRIFT MONITORING	ELEVATED	3 finding(s) · 2 gap(s)

DIMENSION DETAIL

Data Governance — LOW

- ' Data lineage is fully documented and reviewed annually.
- ' Automated data quality checks cover completeness, range, and consistency.
- ' Monthly data quality reports are reviewed by the model owner.
- ' No evidence of data quality validation specifically for the two new engagement features introduced in v3.0.
- ' Lack of documentation on how historical data (2020-2024) handles structural breaks or regime changes in customer behavior.

REG: SR 11-7 Section 4 (Model Development) · SR 11-7 Section 5 (Model Validation)

Model Explainability — LOW

- ' Logistic regression provides full coefficient transparency.
- ' Top 3 churn drivers are delivered in plain language to relationship managers.
- ' Feature importance is documented.
- ' No evidence of user acceptance testing (UAT) to verify that relationship managers correctly interpret the 'top 3 drivers' in complex edge cases.

REG: NIST AI RMF Measure 2.1 (Explainability)

Bias & Fairness

ELEVATED

- ' Disparate impact analysis confirms no differential treatment across demographic segments.
- ' Model does not use protected class proxies.
- ' Bias testing was not explicitly stated for the new v3.0 engagement features; risk of proxy discrimination via engagement behavior (e.g., digital vs. branch usage) is unquantified.
- ' No evidence of intersectional analysis (e.g., age x tenure) in the fairness report.

REG: SR 11-7 Section 5.3 (Model Validation) · OCC 2011-12 (Fair Lending)

Human Oversight

LOW

- ' Model output is strictly advisory.
- ' All retention actions require human approval.
- ' No automated customer-facing decisions or account changes.
- ' No documented training or competency assessment for relationship managers on interpreting model scores.

REG: FFIEC Information Security Handbook

Regulatory Alignment

LOW

- ' SR 11-7 framework applied.
- ' NIST AI RMF governance function referenced.
- ' Clean validation opinion from Q4 2024.
- ' No explicit mention of GDPR/CCPA compliance for data retention of engagement logs.

REG: SR 11-7 · NIST AI RMF

Incident & Drift Monitoring

ELEVATED

- ' Monthly monitoring of AUC, precision, and recall.
- ' Drift threshold defined.
- ' Incident response plan documented for low-risk tier.
- ' No evidence of monitoring for concept drift specific to the new engagement features.
- ' Drift trigger retraining occurred in Q1 2024; no evidence of recent drift analysis post-Q1.

REG: SR 11-7 Section 6 (Model Monitoring)

2LOD CHALLENGE QUESTIONS

Q1 Provide the specific bias testing results for the two new engagement features added in v3.0, including disparate impact analysis across protected classes.

Q2 Demonstrate how the model handles concept drift in engagement behaviors post-Q1 2024, given the recent retraining.

Q3 Show evidence of user acceptance testing (UAT) to confirm relationship managers correctly interpret the 'top 3 churn drivers' in complex, non-linear scenarios.

Q4 Detail the data quality validation process for the new engagement features, including completeness and consistency checks prior to production deployment.

Q5 Provide the training materials and competency assessment results for relationship managers on interpreting model scores and mitigating confirmation bias.

RECOMMENDED ACTIONS

1. Conduct and document bias testing for the new v3.0 engagement features, including intersectional analysis.
2. Implement continuous monitoring for concept drift specific to the new engagement features and report results monthly.
3. Develop and deliver training for relationship managers on model interpretation and bias mitigation, with documented competency assessments.